Sandbar

An exploration towards Software Factory

Why we build Sandbar

What we see

- A person still schedules the agents.
- Chat history is not engineering state.
- An agent sees the task, not the project.
- Coding got faster. Verification did not.
- The engineering base is rented from someone else's model.

What is missing

- General harnesses answer *how to execute*.
- A company must answer *what to execute, and how to know it is done*.
- So three things are missing: durable process state, objective completion, organisational bookkeeping.

Anthropic's Fermat project: about 13 million lines of Lean proof in under two weeks — after a domain harness (Prove2Me) was added.

An amplifier does not produce an organiser. That layer is what we build.

Sandbar develops itself

Bootstrapped with Claude Code (GLM-5.3) for several days. After that, Sandbar builds itself.

HUMANS

- Set the direction: what to build, and in what order.
- Review the design before work starts.
- Sign the merge. The job waits at the gate until then.

AGENTS

- Write the plan, then the code.
- Review the plan; design and run the integration tests.
- Arbitrate, then walk through acceptance.
- Prepare the merge and keep the record.

MODELS IT RUNS ON

role	route
developer	`zai-coding-cn / glm-5.3-flash`
integration	`zai-coding-cn / glm-5.3`
executor	`mimo / mimo-v2.6-flash`

No frontier model. This is not a model demo — it is a process demo.

298 of 1,086 commits changed the engine and its tools. 479 tickets closed across three projects in the month.

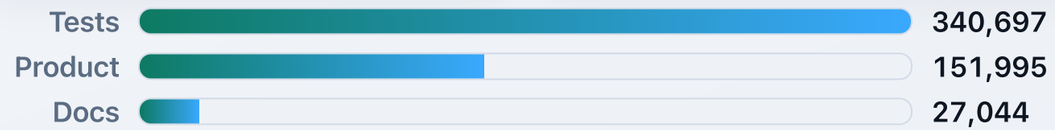
One month. 1,086 commits.

31
days · Sep 10 – Oct 10

1,086
commits

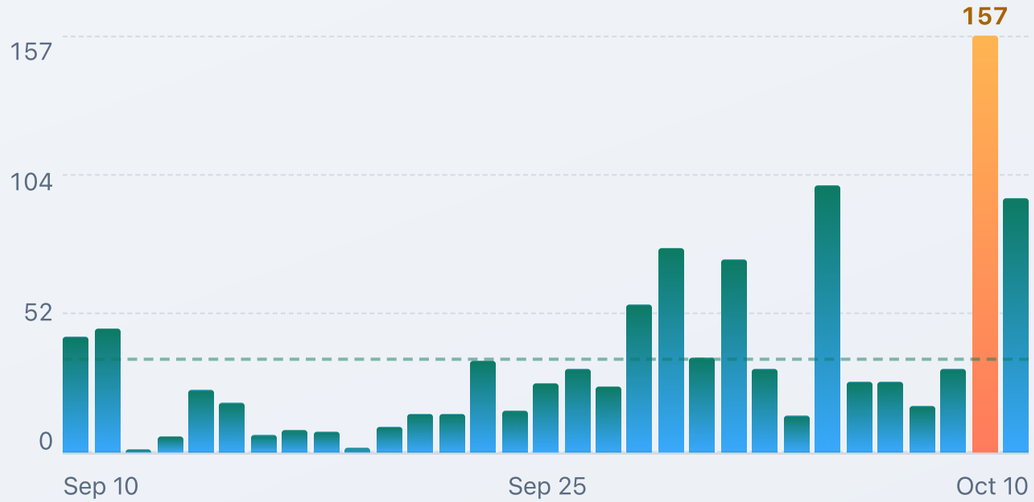
519,736
lines

2,139
files



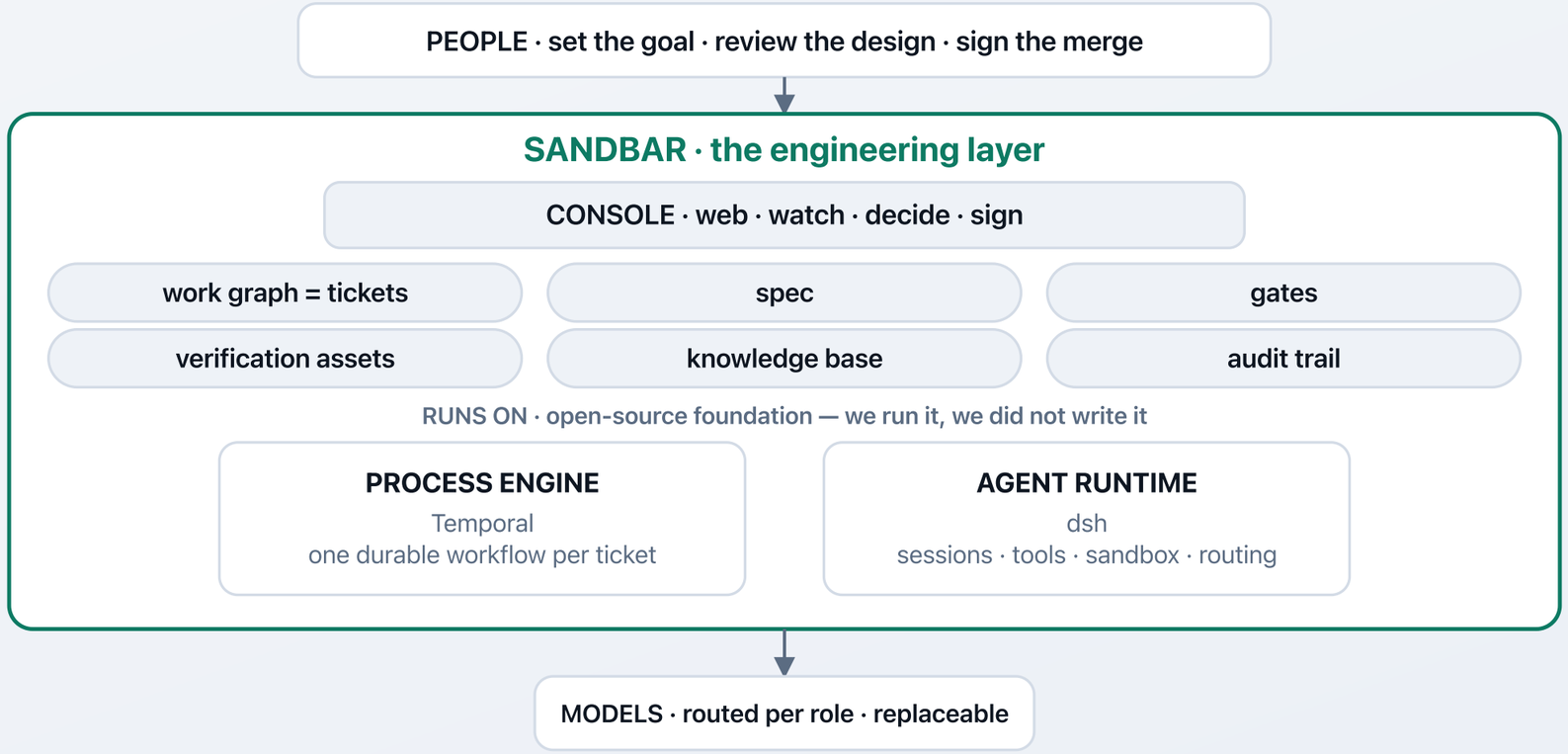
2.2 : 1 test code to product code

Text files only. Generated artifacts (40,085 lines) and binary files are excluded. Lines and files are the repository at the end of the window; source files alone are 461,248 lines.



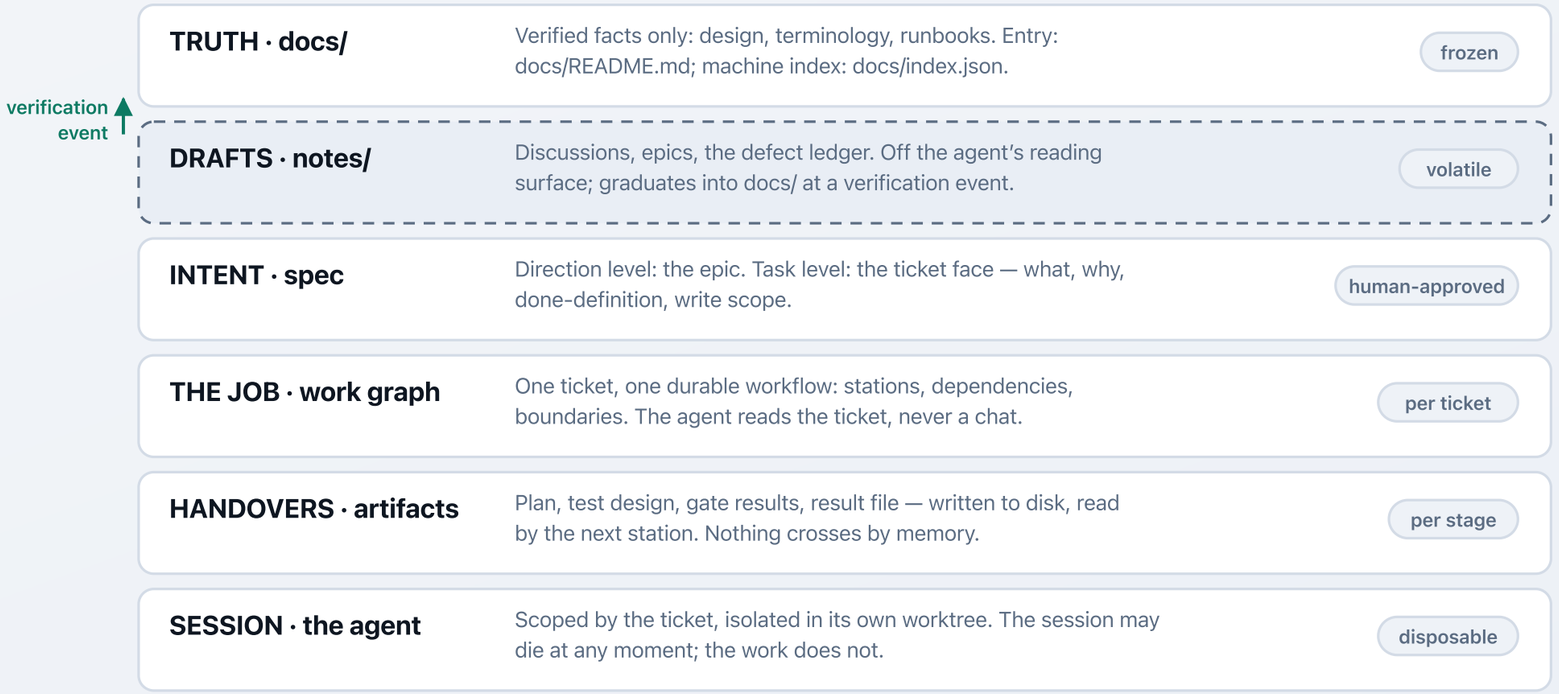
Commits per day, Sep 10 – Oct 10 — 1,086 total, 35 a day.
Dashed line: the average. One day (Sep 12) had no commits.

Engineering IDE of AI era



Nothing inside that box lives in a chat window.

How the context is managed



Trust comes from structure — which directory, which index, which gate — not from asking the model to judge.

Done is decided by evidence

MACHINE-CHECKABLE — script

build · unit · integration · regression guards · performance · merge rehearsal

Anything a machine can decide is never left to a model.

INDEPENDENT JUDGEMENT — agent

another role reviews the plan, designs the tests, arbitrates, walks through acceptance

Whoever produces the work never grades it.

HUMAN SIGNATURE — human

reads the evidence page and the key diff — not two thousand lines; the reason is recorded

The gate physically stops the job until a person signs.

A failed gate sends the work back. Red is named: product, environment, or foreign. Re-running until green is forbidden.

Every fixed defect becomes a permanent guard — the standard of “correct” grows.

The conclusion comes from a judge, not from the player.

Extensibility

THE CHAIN

A project brings its own station chain and loads it at run time.

what you write: a template module

nusuyou: eight stations, two human gates

THE ROLES

A role is a declaration — its rules, its red lines, its handover duties.

what you write: a role file

developer · integration · product-operation

THE MODELS

Per role, per kind of work, per account. No code change.

what you write: a routing table

GLM-5.3-flash for build roles, mimo for the executor

THE PROJECTS

Registering a project is writing one file — nothing else moves.

what you write: one declaration file

three projects on one engine today

THE SURFACE

The console is plugins on a micro-kernel, not a monolith.

what you write: a plugin

board · signatures · metrics · watchdog · providers · knowledge

Nothing about the process is hard-coded: the chain is a template, a role is a file, a model is a table, a project is a declaration.

Extending Sandbar means writing declarations — not forking the engine.

Value: five levers, each with evidence

- **Human leverage** — people appear twice: set the goal, sign the merge. Runs continue unattended overnight.
- **Durable** — sessions die, work does not: a crashed agent retries and continues.
- **Parallel** — isolation, not coordination: three projects, 479 tickets closed in one month.
- **Evidence over review** — read a page of evidence, not 2,000 lines of diff. 268 known defects, each pinned by a regression case.
- **Measurable** — tokens, model cost and stage time per ticket are recorded.

What we have verified, and what we have not

VERIFIED IN DAILY USE

- Three projects, 479 tickets closed in the month.
- Unattended overnight runs; crashed agents recover.
- The engine itself is built this way.
- Every fixed defect has a regression case.

STILL TO MEASURE

- How much orchestration time disappears.
- The real scaling curve past a few parallel jobs.
- A strict A/B: mid-tier model + harness vs frontier model alone.
- Review load, defect escape rate, cross-domain reach.

Biggest risk: a wrong work graph. A bad plan executes very efficiently. Two human gates catch it — a person opens the ticket, and a person signs the merge.

Next: what we want to do

1

Validate on more projects

Today it runs on our own three projects. Take it to eBay projects and prove the pattern is not specific to us.

2

Off this machine, onto the cloud

Engine, process engine and console all run on one box today. Containerise it and make it multi-tenant.

3

Multi-person collaboration

Today one person signs. Several people should work one board, with their own gates and their own audits.

4

Decision models at the gates

Non-generative models that return a choice and a confidence. Testing one on red triage first — it touches no gate before it wins on our own data.

ALSO ON THE LIST · SHARPENING THE VERIFICATION LAYER

- Google Mantis — open-source harness to find and fix bugs; a fix must be proven in a sandbox
- open-code-review — AI review tooling
- mutation testing (Cosmic Ray, MutPy) — would our tests actually catch a planted defect?
- we adopt a practice only when it wins on our own measurements

The ask: continued investment · one real business domain · one team.